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ILTs

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correctness (Heift, &Schulze, 2007). The program does this based on a language model or "expert system" as well as a language corpus, i.e., a large collection of texts in the target language. Advances in the field of artificial intelligence in recent years have resulted in much more successful natural language processing (NLP). Consumers today often encounter NLP, for example, in automated voice exchanges, where NLP is paired with automatic speech recognition, which has likewise improved tremendously.

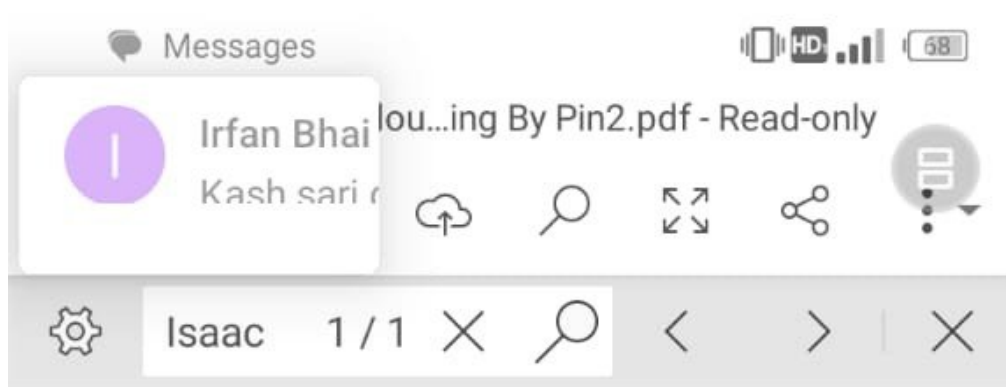
The effort and resources required to build an ILT are such that it would be advantageous for these projects to be built cooperatively or to share resources or code. Most programs are built with computer languages commonly used in computational linguistics, namely Lisp and Prolog, along with those used on web servers, Java and Python, but they are normally built from scratch,

Page 56 of 188

resulting in quite different approaches and code bases. Designing the architecture of the ILT to be modular is a step towards interoperability and flexibility, allowing parts of the system to be turned off or on and to be updated individually. That makes it easier as well to add additional modules in the future. What would increase even more the versatility of ILTs would be the ability for non-programmers to modify the content. The Tutor Assistant is an authoring tool for ILTs which enables that functionality (Toole,Heift, 2002). Flexibility is important as well in providing options to accommodate as wide a range of learners as possible. That could mean providing different pathways for analytical learners who prefer a deductive approach (rules first) and for global learners, who learn better from an inductive method (examples first). When possible, providing different language choices for feedback and other help functions also makes the system more universally usable.

One of the better-known and successful ILTs is E-Tutor for German. Its structure is typical of an ILT in that it incorporates a language knowledge module that parses sentences to provide phrase descriptors and detailed error information, an analysis module, which generates possible responses and updates the student module, which dynamically evolves based on student performance. The nature, amount, and ordering of feedback is one of the most difficult issues that ILT developers face. Language learners will often make more than one error in a sentence so that the system must decide which errors to flag, in which order, and how much feedback to provide for each. Some errors may result from typos, and others may be grammatically incorrect but not critical to comprehensibility. Some systems will provide more detailed feedback to beginners while advanced learners get only a hint at the error.

One approach to user feedback is to have the system places errors into categories and automatically provide feedback on those judged most critical while giving the student the option to see or not feedback on others. Such decisions can be built into the system or be determined by the teacher or by the individual student, by setting defaults in the student module. E-Tutor features an "error priority queue" which ranks errors and is instructor adjustable (Heift, 2004). Another factor for authors to take into consideration in determining user feedback is how much metalinguistic information to provide, that is grammatical or linguistic explanations. Studies



teaching is relatively recent. Except research on Habitat, studies examining VWs have almost entirely been published after 2000.

Given the social nature of VW environments, it is not surprising that one theme shared by several VW studies is their ability to enhance collaboration amongst learners. A second major set of findings regarding VWs and language learning connects to the issue of anxiety. Some of this research found that the use of an avatar (essentially a masked persona for the user) helped learners to “loosen...up a bit...” (Love, Ross, and Wilhelm 2009) while they were able to experiment with new and powerful identities” (Shaffer, Squire, Halverson, & Gee, 2004).

Some researchers also discuss the power of VWs to expose learners to the real world via the virtual environment. While this may seem counterintuitive, it appears that students can become so immersed in the environment that the virtual and the real can become blurred positively.

The Future of VWs

If predictions of future technologies were accurate, contemporary humans would be travelling via personal air cars, with our work being done by groups of robots—made safe, of course, by Isaac Asimov’s three laws of robotics as introduced in the short story Runaround in 1942. Sadly, this is not the case, but near-future predictions about upcoming advances in VWs may be made with greater accuracy based on products currently under development.

Page 47 of 188

Over time, VWs have continued to advance, with massive changes in the quality of the graphics, user interfaces, and possibilities for communication.

However, there are many upcoming changes in these environments that will begin to shift them from being VWs to more fully developed artificial reality simulators (ARS). This begins the shift from a 3D environment on a computer screen to “the idea of immersion—using stereoscopy, gaze-tracking, and other technologies to create the illusion of being inside a computer-generated scene.” (Rheingold, 1991)

Virtual worlds have grown from their text-based infancy of the 1970s to the 3D immersive environments of today. Educators and students now have a wide array of VWs available to them that can grant instant access to native speakers of a huge variety of languages who are located around the world. As these environments and their associated technologies continue to advance, they promise to change the ways that we understand both human-to-human interaction and what it means to be a student of language.

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brief article titled "CARRIE: A Full-Text Online Library," Nelson gives a firsthand account of the site's launch, which he indicates was an immediate success. Within a month the site had approximately three thousand links to historical texts.

Despite its success, CARRIE was soon overtaken by other sites. As Nelson writes, "It was only two months until a new day dawned and colleges and universities throughout the country were scrambling to put up World Wide Web sites, and HNSource and CARRIE lost their uniqueness and were overshadowed by the well-funded and professionally staffed projects that began appearing." Libraries and well-funded projects such as the Library of Congress American Memory project, which grew from a pilot digitization project that ran from 1990 to 1994, were uniquely positioned to develop Internet e-book collections. Aside from their obvious expertise in information management and collection building, they also had the funding and staff necessary to meet the increasing demand for more books, better access models, and more advanced searching tools.

One of the best known of these collections was the University of Virginia Library's Etext Center, founded in 1992. E-book libraries of a similar scale, particularly Internet libraries that offer free e-books, are not common. One that has reached a similar size is the [Internet Archive](#), launched in 1996. The [Internet Archive](#) offers "permanent access for researchers, historians, scholars, and the general public to historical collections that exist in digital format" (www.archive.org). The archive's mission is rooted in the library/archive tradition. It is a visionary effort to create a historical record of a new medium, "to prevent the Internet—a new medium with major historical significance—and other 'born-digital' materials from disappearing into the past." "Born-digital" content included in the archive includes software, audio, video, and e-texts.

Page 70 of 188

As of November 1, 2009, the [Internet Archive's](#) Text Collection contained 1,716,115 items. The largest collection, the American Libraries Collection, included 1,139,936 texts. Other collections are Canadian Libraries, 208,867 items; Universal Library, 70,200 items; and Project Gutenberg, 20,377 items. A list of contributors for the American Collection (www.archive.org/details/americana) shows the diverse institutions involved in digitizing books. Contributors include academic and public libraries, the Boston Library Consortium, Lyrasis, and CARLI, the Consortium of Academic and Research Libraries in Illinois. Corporate contributors include Microsoft, Yahoo! and the Sloan Foundation. Microsoft's contribution alone numbers 339,609 books from its defunct Microsoft Book Search.

The [Internet Archive](#) includes an open-source e-book reader. The reader is developed and maintained by volunteers. In a refreshing move, the [Internet Archive](#) has an open bug reporting system. Readers can report bugs, view a list of known issues, and even see who has been tasked



systems in the 1950s. Eventually, scanners and OCR software would change the role of the early e-book creator from typist to editor/proofreader.

Sharing information on the early Internet was not easy. The Internet, advanced for its time, was still clunky. Download speeds were slow, and basic transfer protocols were in their earliest iterations. TC P/IP (Transmission Control Protocol/Internet Protocol) was developed in 1974, after Hart typed his first e-book. Computer use was not ubiquitous. The first Apple computer, the Apple I, was released in 1977, the first IBM PC in 1981. E-book audiences were small, limited to a handful of academics and scientists.

Early e-book creators faced the twin challenges of building collections and learning how best to use new digital technology. Despite these significant challenges, many of the early collections flourished and survive today.

The World Wide Web and the Expansion of E-books

Many of the technical issues faced by e-books creators were resolved with the advent of the Web, a natural platform for e-books that was far more user friendly than BITNET and previous TCP/IP and FTP platforms. With a viable platform in place and a potential audience of millions, academics, libraries, and enthusiasts jumped into digitization projects and the number of e-book libraries began to increase dramatically. An example of such a site is Renaissance Editions, an online repository of works printed in English between the years 1477 and 1799. The site was founded in 1992 by Risa Bear, a staff member of the University of Oregon Libraries (which hosted the site) and an accomplished poetess with a passion for the literature of the



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Operational help fits into the category of technical support and training. These include user manuals, help menus, training modules, and introductory - level tutorials aimed at making learners aware of how to use the hardware and software, how the program functions, what help options are available and how to invoke them, and potential frustrations.

Page 129 of 188

Regulatory help options are those that aim to prepare the learner for a particular task and understand and reflect on that task after its completion. In the former category, they place listening tips, directions for specific strategy use and development, and guidance on which of the available help options to use for that task. For post - listening guidance, they list explanatory feedback; another example would be directions to reflect or comment on "lessons learned" in line with metacognitive approaches aimed at promoting autonomy.

The provision of compensatory help options taps into the affordances of digital technologies to provide modified input or to make input more salient in ways that can support both comprehension and subsequent acquisition. For modified input, Cardenas - Claros and Gruba note various combinations linking text (subtitles and transcripts) to audio and video, audio to still images, and L1 to L2 transcripts. For increased saliency, they identify the various audio control buttons (volume, rewind, pause), the media controller bar or slider, and for some media players, the variable speed control for slowed playback of audio and video.

Their final category of explanatory help options similarly includes operations targeted at making input more salient through visible links to explanatory hints or links to definitions and glosses from subtitles or transcripts. Explanatory help can also appear in the form of input elaboration, namely, the content found in cultural notes, grammar explanations, electronic dictionaries, and tools such as concordancing programs.

This framework is useful not only for developers. Looking beyond dedicated language learning software with a listening component, language teachers can exploit some of these concepts to help their students have a more productive learning experience with authentic materials. For compensatory help, they can identify specific media players (e.g., the VLC player—see <http://www.videolan.org>) with a rich set of options as well as sources of appropriate online audio or video materials that include captions and/or transcripts. They can offer some regulatory support themselves by suggesting effective procedures for using these materials and media tools. For explanatory help, teachers can also evaluate and recommend electronic



- Technology-mediated listening: Research and practice
- Current trends and future directions

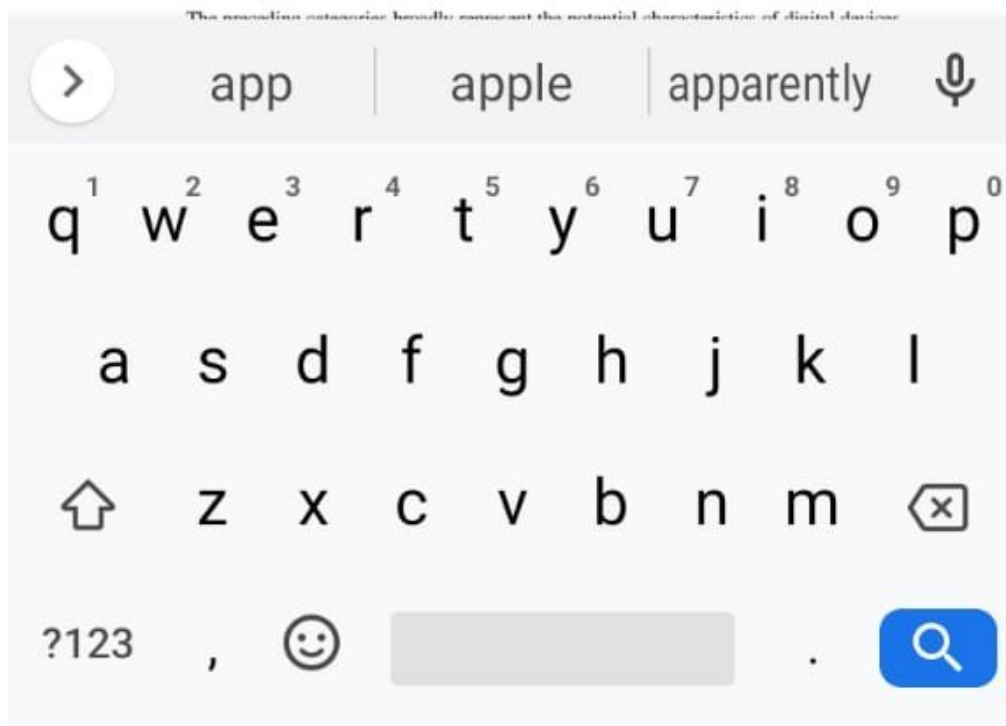
Digital Affordances and New Listening Contexts

In recent years, it has become increasingly common to speak about technology **tools and applications** in terms of their affordances. According to Norman (1988), affordances are the perceived possibilities that a user has for such **tools and applications**. The term is used here in a broad sense where that "perception" is not necessarily universal but rather mediated through theory and practice toward the goals and objectives of second language learning. In this context, referring to the online language learning environment, Lyceum defines affordances as possibilities as well as limitations" having "an impact on its [Lyceum's] use". For example, the affordance of a button for turning on and off captioning in videos may differ depending on whether a teacher or learner perceives it as (1) a shortcut to comprehension that can privilege reading over listening, (2) enrichment of the audio channel to support noticing (Schmidt 1993) during dual coding (Plass and Jones' (2005) multimedia model described below), or (3) a means

to identify and capture new vocabulary items (Danan 2004) in combination with the additional function of pausing.

Technologies for Listening

The preceding categories broadly represent the potential characteristics of digital devices





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classroom. So teachers must maximize the time given to output in the classroom

environment without minimizing the time given to input (such as grammar explanation or clarification, pronunciation practice, etc.) in the classroom because teachers know that the

Page 84 of 188

better the input, the better the output. The beauty of any blended learning course is that the blend can be developed, adapted, changed, tweaked or refined at will, or whenever the paramount needs of the learners demand it. **Choosing the** right blend is a key part of the teacher's role.

Teachers must use the flexibility that blended learning grants them to design the course that best fits learners' needs and wishes whilst taking into account assessment requirements and institutional constraints.

3. Costs

The costs involved in blended learning must be considered as they might not be negligible, especially at the beginning of the process. Cost-effectiveness is a third major goal for blended learning systems. It is undeniable that these provide an opportunity for reaching a large, globally dispersed audience in a short period with consistent, semi-personal content delivery. Some researches have been undertaken in the corporate sector that shows that blended learning which is used effectively provides a large return on investment (Garrison & Kanuka, 2004). However, containing costs requires a fundamental shift in thinking. Rather than focus on how to provide more effective and efficient teaching, we must focus on how to produce more effective and efficient student learning. Once learning becomes the central focus, the important question is how best to use all available resources — teacher time and technology — to achieve certain learning objectives. Using all available resources most efficiently leads to reduced cost and improvement in student learning (Twigg, 2006).

4. Attrition and Motivation

An important element to consider when adopting a blended approach is student attrition and retention. Many students, when they are first introduced to e-learning, tend

to say that they are not 'being taught' and feel that not only have they been wasting their time but that teachers are not doing their job. There are two critical elements at play here:



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today, language courseware is overwhelmingly designed to be delivered over the internet. In the 1990s, most of that development was done in programming environments that were not native to web browsers namely Java and Flash and which necessitated the use of special browser plug-ins to run. Plug-ins are software components added to web browsers that enable special features or functionality. Several ILTs, including E-Tutor, are written in Java and designed to run in the Java plug-in as applets in a web browser. Flash has also been used extensively, for example, for the rich internet applications from the Center for Language Education and Research (CLEAR, Michigan State University). The use of Java and Flash enabled the kind of interactivity and media integration that at that time were not possible in the native web environment using HTML. Plug-ins, however, were not an ideal solution, as they tended to cause performance issues and also could not be fully integrated into web pages. Another early option to move beyond static webpages was server-based interactivity through CGI scripts, or common gateway interface, written in Java or Perl. This method however could be slow, as a new page had to be received from the Web server, and since the entire page was replaced, did not work well for designing interactive courseware. The arrival of JavaScript in the mid-1990s changed dramatically the nature of Web interactivity.

JavaScript (officially ECMAScript) is a scripting language that is incorporated into the source of the webpage, along with HTML and CSS. JavaScript is client-side code and is natively supported by Web browsers, which means that it runs efficiently and can be integrated into the Web page structure. What that means in practice is that a script can manipulate objects on the page, for example, showing a checkmark for the correct answer to a question and providing appropriate feedback.



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different geographical sites can easily collaborate on a VoiceThread project. When VoiceThread slides are shared online, the creators can allow anyone with an Internet address (URL) to add comments or answer questions posed by the creators, thereby enriching the entire project. The

creator/instructor can always maintain control over which comments are posted by choosing the < moderate comments > option under the publishing menu. Because VoiceThread permits the formation of learning groups at a distance, it might also be considered a form of asynchronousCMC, but we have included it in tutorial CALL because individuals can always develop single - authored presentations on their own in a completely independent fashion. In practice, VoiceThread can be used to conduct both tutorial CALL for speaking or asynchronous CMC.

Page 155 of 188

Learners should use the following storytelling tools:

- **Omeka** : Open or Hosted resource for archiving and displaying images and content. Free
- **Timeline.is** : Using Java and a Google Sheets template, you can quickly create interactive timelines with images, videos and other content.
- **We Video** : A simple web based story editor
- **Neatline** : Mapping and Timeline Tool for Omeka Projects.
- **ArcGIS Online** : University of Minnesota ArcGISOnline Server access.
- **Census Data Viz** : Examples of data visualizations from Census Data.
- **Canva** : Freemium graphic design tool with infographic templates.

Digital Sound Tools: Unleashing Student Creativity

In addition to the applications presented above, L2 students have at their fingertips many digital tools that can easily be used to embellish their own stories with images and sound to produce multimedia materials for the Web.3 Blake and Shiri (2012) described how the first year Arabic students at UC Berkeley demonstrated their speaking progress at the end of the term by publishing their audio - enhanced finished products on YouTube.com. Several groups of students in this Arabic class opted for simple hand illustrations with hand - written Arabic captions, adding in later their audio narration (<http://www.youtube.com/watch?v=Gc4gw6Ig564>). These easy - to - use techniques result in an impressive finale to a language course to the great satisfaction to both students and instructors alike and can be viewed by the entire class to provide feedback on the students' L2 speaking.

We recommend some authentic video sources to use in the ESL classroom.

- University channels (Berkeley, Harvard, Stanford, Oxford etc)

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learning activities is focus on form (Font, Long, & Robinson, 1999) where the breakdowns, once again, give rise to an examination of new linguistic structures that remain to be learned and controlled by the L2 learner.

Still other researchers, such as Ellis and Sheen (2006), have focused more specifically on the feedback learners received when breakdowns occur, which includes linguistic recasts or reformulations that more proficient partners provide L2 learners in response to the ongoing

conversational exchanges. As part of this feedback routine, Swain (2000) also included the L2 speaker's feedback loop that arises when pressed to produce output during a conversation; in other words, at the moment of speaking, the L2 learner must try out specific grammatical forms

and lexical items, which in turn, results in felicitous or infelicitous utterances. The process of trying something out can be a point of learning, particularly if the speaker notices his or her lack of knowledge and then discovers how to modify and improve the original utterance.

Autonomy and Student Agency

In general, current constructionist educational approaches endorse a more student-centred learning environment over a traditional teacher-centred model. Accordingly, many researchers have focused attention on creating an atmosphere where the principles of student autonomy and self - agency are heightened (Little 2007), which is especially relevant for the CALL context (Blin, 2004; Guillen, 2014; O'Dowd, 2007; Schwienhorst, 2008). Briefly, autonomous learning means that students take responsibility for their learning process, which should not be confused with self - instruction. This approach emphasizes learner empowerment, language reflection, and, finally, appropriate L2 usage. The teacher is involved at all stages as a guide, whereas with self - instruction there is no instructor presence at any point.

Page 148 of 188

In the CALL context, increased student agency not only guides how students use technology but also implies that they become creators of L2 texts through their active creation of blogs, wiki entries, chat exchanges, audio and video postings (Blake 2013). Some blog creation sites have been shared with you.

- **Blogger** (<http://www.blogger.com>): It is a free blog publishing tool from Google for easily sharing your thoughts with the world. *Blogger* makes it simple to post text, photos and video.
- **WordPress** (<http://www.wordpress.com>): It is web software you can use to create a beautiful website or blog.

- **Blip** (<http://www.blip.com>): It provides a fully featured publishing platform for text, photos, and video.

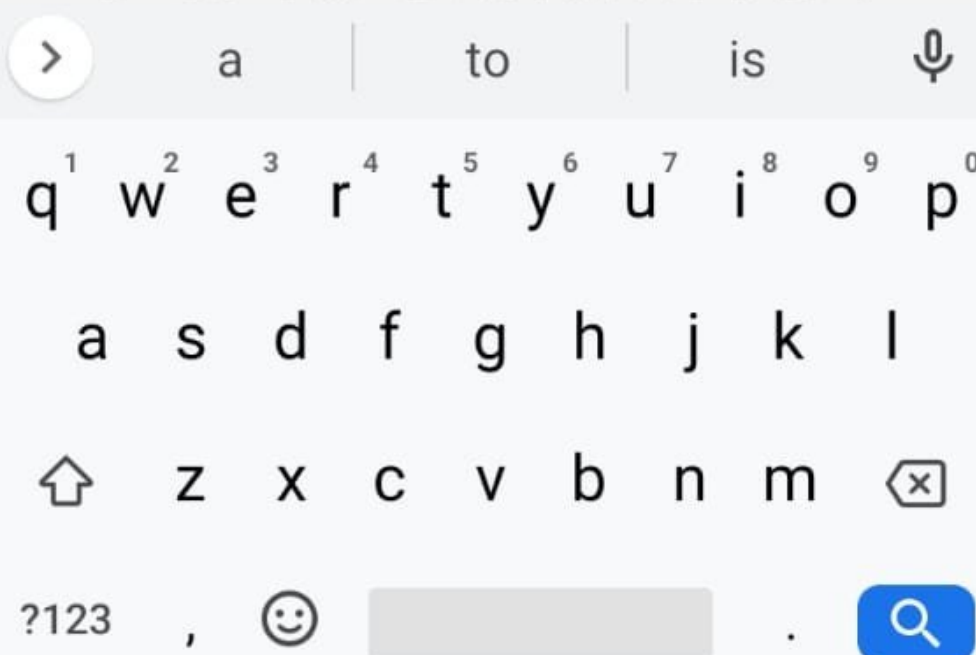


(Rayner, 1998). The duration of an eye fixation is interpreted as an indicator of the processing time applied to the object being fixated on. A long fixation duration indicates either difficulty in extracting information, or greater engagement with the object (Just, & Carpenter, 1976).

Smith (2010, 2012) investigated the use of eye-tracking technology to help pinpoint learners' attention to and noticing of feedback in an L2 SCMC task-based environment. Recasts were chosen as the specific type of corrective feedback (CFB) investigated. Using eye fixation data as a measure of "noticing events" learners were found to attend about 60% of the **intensive recasts** they received by a native speaker interlocutor. Lexical recasts were much easier than grammatical recasts for learners to not **accurately** produce on subsequent chat interaction and delayed written post-tests.

Using heat map records and stimulated recall records, Smith (2012) found that both stimulated recall and eye-tracking records were favourable predictors of learner noticing with the eye-tracking record being slightly more discriminating in its prediction on immediate posttest success. There was no significant relationship between the linguistic category of the recast and noticing in the eye-tracking condition. However, in the stimulated recall condition learners were much more likely to report having noticed recasts targeting semantic features than morphological recasts. This last finding has implications for SLA as it suggests that learners show evidence of heightened attention to recasts, regardless of the linguistic category of the

target. When considered in combination with the results from Smith (2010) these findings suggest that even with heightened attention on morphological recasts, learners were less able to attend to these features at a level that allowed them to make use of the teacher recast on the post-test. For SLA theory this is important as it is consistent with previous research by Mackey,





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theoretical rationale for hypothesizing that the affordances of a specific CALL environment are likely to impact the results at a process or outcome level (question 4 above). For example, in an examination of explicit recasts in an SCMC environment, Smith (2012) argues that such written corrective recasts are necessarily more explicit (and therefore more noticeable) than those found in oral recasts due to the written nature of the medium itself as well as the message permanence afforded.

Three SLA - relevant CALL studies

Using the four considerations discussed above, I outline three exemplary SLA - relevant CALL studies from three of the most widely employed theoretical frameworks. Each of these studies focuses on computer - mediated communication in one form or another, reflecting CMC's predominant position in CALL research in recent years.

➤ Cognitive Approach

Payne and Whitney (2002) used Levelt's (1989) model of language production and concepts from Working Memory theory (Baddeley, 1986) to explore whether SCMC can help

improve learners' L2 oral proficiency by developing the same cognitive mechanisms that underlie spontaneous conversational speech.

Is the study primarily concerned with employing computer technology for teaching and learning foreign languages?

Four sections of a third semester university - level Spanish course were used for this study.

These were intact groups with the students in the experimental groups (meeting online in a chat room) engaged in the same activities on the same days as those in the control groups did face to



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Page 171 of 188

group) would show at least the same degree of oral proficiency development over one semester as those participants in the control group. In terms of research question 2, there was a lower correlation between oral proficiency gain scores and one of the WMC tests in the experimental condition (SCMC) as compared with the control condition (F2F), indicating that learners with a lower WMC benefited from the chat environment.

➤ Interactionist approach

In a study of teacher - learner (text - based) SCMC interaction during online conferences, **Smith and Renaud (2013)** used eye - tracking technology to explore the relationship between [redacted] of the recast targets.

Is the study primarily concerned with employing computer technology for teaching and learning foreign languages?

This study examined the effectiveness of teacher - provided corrective feedback during one-on-one online conference with students. The task was for students to meet virtually with the instructor and discuss (in the target language) a draft of a written assignment due the next week. The teachers were asked to provide full recasts to learners when it seemed natural to do so. Recasts may be defined as discourse moves by an interlocutor that, within the context of communicative activity, correctly restate or rephrase all or part of a learner's utterance to be

more target - like by changing one or more sentence components, while still acknowledging and retaining its central meaning. For example, when a learner of German says something like Ich habe Hunger nicht* (I am not hungry) the teacher may provide the recast Ah, Sie haben keinen Hunger (Oh, you're not hungry). The online conferences were viewed by the instructors as part of the regular curriculum, and they typically held either face - to - face or online conferences



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- Digital Books
- Classroom Clickers
- Interactive Whiteboards and Tablets

Integrating Technology

The integration of technology should serve to guide, expand, and enhance learning objectives. It is understood that teachers and students do need to spend time learning the basics of using a computer. This is necessary to move to effectively integrating technology. Curriculum integration with the use of technology involves the infusion of technology as a tool to enhance learning in a content area or setting. Technology enables students to learn in ways not previously possible. Effective integration of technology is achieved when students can select technology tools to help them obtain information promptly, analyze and synthesize the information, and present it professionally. The technology should become an integral part of how the classroom functions as accessible as all other classroom tools.

Curriculum Design

Curriculum design is a term used to describe the purposeful, deliberate, and systematic organization of curriculum within a class or course. In other words, it is a way for teachers to plan instruction. When teachers design a curriculum, they identify what will be done, who will do it, and what schedule to follow.

It is generally understood as a high-level process defining the learning to take place within a specific program of study, leading to a specific unit(s) of credit or qualification.

Teachers design each curriculum with a specific educational purpose in mind. The



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2015). Naturally, the instructor must prepare the conversational tasks ahead of time so that the

students know exactly what to expect and are primed with the appropriate vocabulary and grammar constructions needed to bring the task successfully to completion. In this way, students can gradually build up greater fluency and smooth discourse transitions, which are important components of speaking proficiency.

Unfortunately, speaking progress via videoconference usage has not been well tracked or measured to date, partly because defining the construct of speaking proficiency remains a daunting, if not insurmountable, undertaking (Hulstijn, 2011) whether in the face - to - face or CALL context. Hulstijn (2015) has recently challenged researchers in the L2 field to test out empirically whether or not speaking proficiency should be conceived as a continuum or, as he has suggested, a dichotomy (i.e., basic vs. extended linguistic proficiency). These same L2 theoretical issues are relevant to CALL research, as well, and deserve serious attention in future CALL research.

Chatbots have revolutionized academia as well as business. Amazon and other international e-commerce tycoons use these chatbots. [Dialogflow](#), a platform supported by

Page 160 of 188

Google, can train chatbot without any coding skill, even students of humanities have modified and built [chatbots from scratch](#). These chatbots have been designed with variety of technology, diversified instances in different languages of the world for instance Babbel, Rosetta Stone, Duolingo, Memrise, Andy, Mondly etc

In the course of discussing SCMC, we do not wish to imply that videoconferences are



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well as what the researchers count as data, and accept as evidence. This fact makes it difficult to talk about SLA as a cohesive entity. Accordingly, we frame our discussion of technology-enhanced SLA research in terms of three theoretical frameworks that are widely researched and especially compatible with CALL inquiry: cognitive approaches, the interaction approach, and sociocultural theory.

➤ Cognitive Approaches to SLA

There are a host of cognitive approaches to SLA, which can be broadly classified into traditional information processing and emergentism (Ortega, 2009). Cognitivists believe that to properly understand the final state of fluent L2 development we need to understand the processes by which this expertise came about. Cognitive science is also concerned with functional and neurobiological descriptions of the learning processes, which, through exposure to representative experience, result in change, development, and the emergence of knowledge (N. C. Ellis, 2002). Information processing theories all share the belief that human cognitive architecture is **made up of** representations and access. Mental processing is comprised of automatic (unconscious) **processes** and voluntary or controlled (conscious) operations. Cognitive resources such as

attention and memory are limited, and performance is variable and vulnerable to stressors, such as those factors that make a language learning task more difficult such as time pressure.

In contrast, emergentist SLA scholars argue for a usage-based explanation of language development (including SLA). They argue that fluent language use is exemplar-based and is **made up of** "a huge collection of memories of previously experienced utterances" rather than knowledge of abstract rules (the latter explanation championed by generative linguistics). Emergentists take most language learning to be "the gradual strengthening of associations between co-occurring elements of the language," and they see fluent language performance as "the exploitation of this probabilistic knowledge" (Ellis, 2002, 173). Cognitive theory directs researchers to investigate constructs such as working memory, which can be assessed through the use of technology.



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Computers can help teachers and students find old information and new information from thousands of sites on the Internet that are updated daily. Computers enable students to view live, synchronous events happening almost anywhere in the world or from space (e.g. the current weather as seen from a satellite). Computers can be used to manipulate data so that students can grasp quickly how a change in one variable in a system affects other variables. Computers enable students to assemble and create new information. Computers enable students to have text, audio, and graphic information. They have access to in some less densely populated areas. Computers allow students to communicate with just about anyone, anywhere, and anytime. (Valmont and Wepner, 2000)

On the other side of the coin, computers cannot replace teachers. Valmont and Wepner quote in 2000:

"Contrary to the wisdom of the most fanatical technologists, computers do not think, initiate, or react the way teachers do. Although some people believe that information equals education, teachers know that understanding is not automatic with the acquisition or memorization of facts. Knowing how to apply information, how to use ideas in new ways, how to evaluate information, and how to extrapolate or go beyond basic information are important aspects of learning."

Teachers understand when the teachable moment is at hand and guide students' academic growth. The teacher facilitates learning and helps students transfer ideas to new learning

situations. Integrating technology isn't about using complex technology programs but rather simplifying technology choices and focusing on how technology connects to learning.



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**Benefits of Using Technology in the Classroom****Improves Engagement**

When technology is integrated into lessons, students are expected to be more interested in the subjects they are studying. Technology provides different opportunities to make learning more fun and enjoyable in terms of teaching the same things in new ways. For instance, delivering teaching through gamification, taking students on virtual field trips, and using other online learning resources. What is more, technology can encourage more active participation in the learning process which can be hard to achieve through a traditional lecture environment.

Improves Knowledge Retention

Students who are engaged and interested in things they are studying are expected to have better knowledge retention. As mentioned before, technology can help to encourage active participation in the classroom which also is a very important factor for increased knowledge retention. Different forms of technology can be used to experiment with and decide **what works** best for students in terms of retaining their knowledge.

Encourages Individual Learning

Page 92 of 188

No one learns the same way because of different learning styles and different abilities. Technology provides great opportunities for making learning more effective for everyone with different needs. For example, students can learn at their speed, review difficult concepts or skip ahead if they need to. What is more, technology can provide more opportunities for struggling or



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d. student response systems

a. **Tests, quizzes, and exams** taken on a computer screen under the supervision of an invigilator are known as **online tests, quizzes, and exams**. The tests and exam questions are already created by the instructors and uploaded online at the university/college's website or canvas.

b. **e-submission** is the process by which a student submits work online for storage and retrieval by academic, administrative or other staff for marking, feedback, or review. This can also include originality checking to help identify plagiarism using Turnitin.

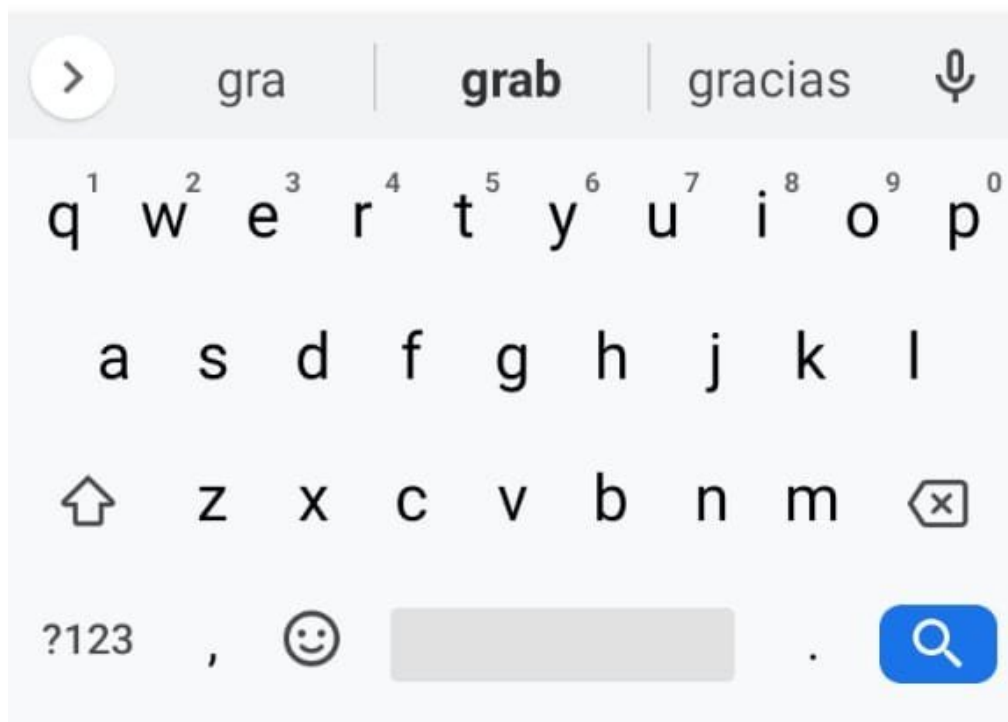
e-marking is the process by which staff provide marks and feedback to students online, which can include online annotations e.g. using Blackboard inline grading, annotated files e.g. using comments in Microsoft Word, audio, video, or the use of online forms.

c. **Self-assessment** means the process of having the learners critically reflect upon, record the progress of and perhaps **suggest grades** for, their learning.

The term **peer-assessment** refers to the process of having the learners critically reflect upon, and perhaps **suggest grades** for, the learning of their peers.

Reasons for Using Self- and Peer-assessment

Self-assessment





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By using metacognitive strategies, readers are better at activating background knowledge, identifying and planning reading tasks, and monitoring their reading (Boulware-Gooden et al., 2007).

SLA Theories and Applications of Technology to L2 Reading

When reading is seen as a means for second language acquisition and technology is added, the L2 reading process becomes even more complex to conceptualize. Digital environments add visual and auditory features to a text and require the reader to integrate the information afforded by these modes into the processes of comprehension. Furthermore, with the interactivity provided by Web technologies, reading is supplemented by a social dimension of instantaneous connectedness not just with text but also with other people for exchanges and co-construction of understanding. To create and evaluate technology for language learning, computer-assisted language learning (CALL) developers and researchers have looked to second language acquisition (SLA) theories for support and guiding principles. Similar to reading theories, SLA theories have also gone through stages of transition. Although SLA approaches of

the different stages provide distinct implications for CALL, cognitive, linguistic and psycholinguistic approaches have been most influential for the study of applications of technology to L2 reading (Chapelle, 2009; Chun, 2011a). More recently, as the Internet provides far-reaching and instantaneous interactivity, researchers and educators increasingly rely on sociocultural approaches.

Page 103 of 188

The focus of cognitive theory on noticing, working memory, automatization of word recognition, and activating prior knowledge has provided theoretical bases for many studies



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Page 100 of 188

- SLA theories and applications of technology to L2 reading
- Technologies in use for teaching and learning of L2 reading
- Self-developed and commercial courseware
- CMC technologies (Chat, Moo, email)
- Challenges
- Future directions

Introduction

With widespread access to reading resources and various forms of reading, the concept of reading nowadays may be different from what it once was. Online reading allows readers to interact in compelling ways with text and with others, but readers are at the same time being exposed to the danger of distractions or are overloaded by the volume of materials (Godwin-Jones 2010). This can be especially true for L2 learners reading in a language that they are not proficient in because they need support to overcome linguistic and processing practice limitations (Cobb, in press; Grabe 2014).

Simultaneously, many Web 2.0 tools and technologies are being designed and used for all levels of L2 learners to build certain skills for effective L2 reading. L2 educators play vital roles in helping students acquire the skills necessary to take advantage of new technological tools so they can actively participate in global communities (Gonglewski and Dubravac 2006). The objectives of this chapter are:

- (1) To inform L2 teachers, researchers, and software designers of the theoretical grounding of applying technological innovations to the teaching of L2 reading
- (2) To introduce the variety of technology applications to foster successful L2 reading
- (3) To discuss the pitfalls and challenges of reading in the technology age
- (4) To provide thoughts on future directions for applying technology to L2 reading development and the possibilities of cultivating ubiquitous, learner-centred, autonomous learning.

L2 Reading Theories



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examination at the same time, as well as the need to ensure a stringent level of security is among the concerns that need to resolve to accomplish this transition.

E-marking is one way that many exam assessments and awarding bodies, such as Cambridge International Examinations, are utilizing innovations in technology to expedite the marking of examinations. In some cases, e-marking can be combined with electronic

Page 97 of 188

examinations, whilst in other cases, students **will still** handwrite their exams with papers scanned in securely and uploaded into an e-marking system.

Different Forms of E-assessment

Some of the most widely-used forms of e-assessment are:

- a. online tests, quizzes, and exams
 - b. e-submission and e-marking
 - c. self- and peer-assessment
 - d. student response systems
- a. **Tests, quizzes, and exams** taken on a computer screen under the supervision of an invigilator are known as **online tests, quizzes, and exams**. The tests and exam questions are already created by the instructors and uploaded online at the university/college's website or canvas.
- b. **e-submission** is the process by which a student submits work online for storage and retrieval by academic, administrative or other staff for marking, feedback, or review. This



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The most general purpose of the three is the Web 2.0 application, which allows for authorship of multiple users who contribute content to a common repository online. The second category is automated writing evaluation systems developed specifically for analysis and feedback to writers during the writing process. The third category is the corpus - based tools, which serve as a reference to writers as they strategically examine the language used in collections of existing electronic texts. Many software tools exemplify each of these categories, but we have chosen a few to illustrate the affordances offered writers by tools of each of the three types.

➤ Web 2.0 Applications

Web 2.0 tools include a variety of social media sites, such as Facebook and Twitter, as well as blogs. For L2 learners, these sites provide unprecedented opportunities to experiment with their language in settings where their language appears before and communicates with real audiences rather than being confined to communication for practice within the language classroom. The use of language for communication, rather than for practice alone, is a **core goal** of most language instruction. Beyond opportunities for communication, Web 2.0 applications like Google Docs provide the tools for collaborative writing.

Lang - 8 (<http://lang-8.com/>) is a good example of a social networking site designed to encourage L2 communication by creating a community where language learners are matched with native speakers of target languages. Each member can contribute to the virtual language learning community by writing posts in the language they are learning and giving feedback or making corrections to other members' written posts in his or her native or stronger language.

With a few buttons for editing, such as colour changing and crossing out, users can easily correct or edit others' posts on Lang - 8. Lang - 8 rewards active users with "L" points, which help make their posts more visible to proficient users, thus increasing the probability that they will receive feedback or corrections. In other words, the more a member of the community contributes to the process of language correction, the more his or her writing or posts are corrected. In addition to written corrections, Lang - 8 has integrated Skype, an online tool allowing for international, live video communication. Lang - 8 allows learners to practice



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basic and effectiveness of technology-mediated distance language teaching, arguing with advancing theory, research and practice to inform what a burgeoning domain of activity is.

Defining features of all these forms of distance language teaching are that teachers and learners are physically separated, that "the bulk of the learning takes place in non-co-presence" (Lamy 2013a, 144), and that technology is used to mediate the teaching-learning processes within the presence of an educational organization (distinguishing it from the private study). Distance language teaching has evolved largely in response to developments in technology, from early print-based courses, to educational radio and broadcast television, through to audio and videocassettes, and then to computer technologies, with further possibilities offered by interactive multimedia, Web 2.0 environments, and Second Life. Developments in technology-mediated distance language teaching mark a fundamental shift from early approaches concerned with the production and distribution of learning materials for independent study (including CD-ROMs, video-based courses, and broadcast education for example), to more contemporary approaches concerned with interaction, communication, collaboration and collective activity within virtual learning environments. Technology is no longer solely used for distribution purposes, as in broadcasting, and emphasis is now placed on opportunities for communication using both text and sound, as in the chat, and videoconferencing, meaning that distance language education **can focus** on communication and learning as a social process.

Importantly, too, the new learning environment offers opportunities for peer support, and for reflection on learning experiences in both private (with teachers) and shared (with peers) conversations online. Thus, emerging paradigms for distance language teaching have made possible different combinations of individual and collaborative language learning environments and called for not only technological innovation but pedagogical innovation. They introduced new expectations of what is required to work successfully in technology-mediated distance language learning environments for both learners and teachers. Earlier research using older technologies such as the telephone (Graham 2000) can acquire new significance with the introduction of accessible tools such as Skype; the value of those earlier studies often lies in the



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In contrast to the proliferation of multimedia CD-ROMs for language learning, there have been relatively few publicly released ILTs. A good number of projects received preliminary funding to support a proof-of-concept prototype but were not able to fund final development to enable production versions. This is not surprising, given how difficult a task ILTs face, dealing with the complexity of human language and of second language acquisition. ILTs often are designed as research demonstration projects, rather than shareable or commercial products. There are relatively few reviews or studies of ILTs with the exception of those by the developers themselves. Developers of ILT tend to focus, as one would expect, on the functionality and effectiveness of the software, rather than on the user interface design. The user experience can be

Page 60 of 188

quite different from that provided by commercial software. In particular, graphics and multimedia do not display the same high production values.

Commercial Courseware and Open Educational Resources (OER)

In recent years, language-learning software such as **Rosetta Stone** or **Tell Me More** has gained prominence (Nelson 2011). Although such products are sometimes used to provide the primary delivery system for language courses, they are more often used as supplements to classroom instruction or independently by learners in self-instructional contexts. Integrating such commercial software into instructed language learning can be problematic in terms of vocabulary, grammar sequencing, and cultural content. A commercial alternative to standalone software such as Rosetta Stone are the quite widely used electronic workbooks which many publishers are now supplying with basic language textbooks. WWW.duolingo.co is a great source to learn any foreign language. It starts with GTM (Grammar Translation Method), and afterwards, it shifts to the direct method. It is entirely free, and language learning has become a fun with audio visual aids.

Unfortunately, there have been a few studies on the effectiveness of publishers' electronic workbooks. The use of such resources, given the expense and comprehensiveness, is likely to mean that local resources are not developed in support of language instruction. That may not be an issue in some educational settings, but it is possible that curricular requirements or standardized testing may make it imperative to be able to modify materials to meet local needs.

Given the expense, limited access, and inflexibility of much commercial software, one of the recent developments in the creation of language courseware is the rise of shareable teaching and learning materials, often referred to as open educational resources (OER). These can range from quite simple and basic, such as vocabulary grammar drills created with tools such as Quia (a web-based exercise creator) to full-fledged courseware incorporating grammar tutorials, exercises, readings, dialogue, and multimedia.

The well-regarded language courses available from the BBC also are in this category, as



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others.

Page 46 of 188

• **Education focus (low, mid, or high):** Nearly all VWs have potential use for education. The key is in the way that they are used. VWs are a tool for teachers in the same way that PowerPoint can be a tool.

• **Technical requirements (low, mid, or high):** Some VWs are more graphics-intensive than others. Some require more bandwidth, or are more difficult to navigate.

Research on Virtual Worlds

In comparison to other forms of computer-mediated communication (CMC) such as email, message boards, and so on, research into the effect of VWs on language learning and teaching is relatively recent. Except research on Habitat, studies examining VWs have almost entirely been published after 2000.

Given the social nature of VW environments, it is not surprising that one theme shared by several VW studies is their ability to enhance collaboration amongst learners. A second major set of findings regarding VWs and language learning connects to the issue of anxiety. Some of this research found that the use of an avatar (essentially a masked persona for the user) helped learners to "loosen...up a bit..." (Love, Ross, and Wilhelm 2009) while they were able to experiment with new and powerful identities" (Shaffer, Squire, Halverson, & Gee, 2004).

Some researchers also discuss the power of VWs to expose learners to the real world via the virtual environment. While this may seem counterintuitive, it appears that students can become so immersed in the environment that the virtual and the real can become blurred positively.

The Future of VWs

If predictions of future technologies were accurate, contemporary humans would be travelling via personal air cars, with our work being done by groups of robots—made safe, of course, by Isaac Asimov's three laws of robotics as introduced in the short story Runaround in 1942. Sadly, this is not the case, but near-future predictions about upcoming advances in VWs may be made with greater accuracy based on products currently under development.



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for CMC, synchronous and asynchronous, in text, oral, and visual format. Chat, for example, has played a crucial role in providing interactive opportunities in distance language courses: Volle (2005), for example, details synchronous online oral tasks and online oral interviews incorporated into a course for distance learners of Spanish, which was then extended into desktop videoconferencing.

Asynchronous forums were among the earliest forms of CMC used in distance language teaching, with attention to ways in which reflection could be optimally combined with interaction in text - based conferencing. Subsequently, both asynchronous and synchronous opportunities were used together as in the use of email and chat in an online distance writing course (Raskin 2001), at which point access to the opportunities needed to take account of time zones. A new issue was that of vicarious interaction, that is learners who preferred to read rather than contribute to online interactions. Importantly, this led to new understandings of the importance of social presence online, and the need to integrate computer - mediated communication into the course curriculum and assessment.

CMC is important in distance language teaching not only as a learning tool but as a gateway to target language communities.

- Audiographic and Videoconferencing Environments



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reflective interaction among distance language learners of French in a project termed Lexica Online. The project aimed to encourage student reflection on their vocabulary learning strategies through group discussions which were moderated by tutors over six weeks. A significant feature of the study is that it analyzes the contributions of tutors and students in terms of message types and tutor styles, both of which are linked to student learning. Of the three text types, only "reflective conversations" (as opposed to monologues and social conversations) were identified as contributing significantly to language learning in that they were interactional "in both information processing - and social - interactional senses". In terms of optimal tutor styles,

Page 40 of 188

Lamy and Goodfellow argue the need to attend to both cognitive and social dimensions of student participation and learning. Their work was part of a wider concern as to how to develop learner autonomy within collaborative online distance learning as a critical dimension of learner support.

Hampel (2009) identifies a key challenge in online distance synchronous settings as addressing the tendency for tutors to assume a more directive or teaching-centered approach than is congruent with the espoused benefits of learner-centered language learning. In fact, certain features of the setting may lead to this approach including student unfamiliarity with the online tools, technical problems, and the perceived need to direct students to the affordances of the environment. Importantly too, Hampel notes that students often find it very difficult to collaborate, having been mostly used to the self-study aspects of distance language learning. Beyond these epistemological shifts in practices of distance language learning and teaching, Hampel identifies the critical importance of design in fostering interaction and collaboration online, referring to Mangenot and Nissen's (2006) framework: "A more learner-centered approach requires the ability on the part of the teacher to provide a setting in which learners can develop the socioaffective, sociocognitive and organizational skills that are prerequisites of collaboration. This can be facilitated by appropriate tasks, moderation and feedback."

These findings underline the importance of affective and organizational dimensions of engagement across all aspects of the role: students for example argued that for their points of contact with tutors were generally high stakes, and that distance language teaching thus requires more attention to interpersonal aspects and relationships, better organization and focus than was often required in other settings, and a degree of sensitivity and empathy towards the learner's individual context. Much research has focused on understanding distance teacher skills in different virtual settings and concerning specific task types (e.g., Comas-Quinn, de los Arcos, & Mardomingo, 2012; Ernest, Heiser, & Murphy, 2013), with a sustained focus on how to encourage interaction and collaboration, to maintain motivation and to ensure student retention and course completion. Important operational practices include setting up online socialization in multimodal learning environments, providing access to support and timely feedback, ensuring congruence between coursework and course assessment, providing spaces where students can



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small, limited to a handful of academics and scientists.

Early e-book creators faced the twin challenges of building collections and learning how best to use new digital technology. Despite these significant challenges, many of the early collections flourished and survive today.

The World Wide Web and the Expansion of E-books

Many of the technical issues faced by e-book creators were resolved with the advent of the Web, a natural platform for e-books that was far more user friendly than BITNET and previous TCP/IP and FTP platforms. With a viable platform in place and a potential audience of millions, academics, libraries, and enthusiasts jumped into digitization projects and the number of e-book libraries began to increase dramatically. An example of such a site is Renaissance Editions, an online repository of works printed in English between the years 1477 and 1799. The site was founded in 1992 by Risa Bear, a staff member of the University of Oregon Libraries (which hosted the site) and an accomplished poetess with a passion for the literature of the period. From 1992 to 2005, Bear and her contributing editors published over 164 e-books on the Web, including works by Shakespeare, Francis Bacon, Mary Wollstonecraft, Edmund Spenser, and Jonathan Swift.

As these libraries grew, people with similar interests began to collaborate and share ideas. E-book communities began to emerge and libraries began to merge. The Luminarium was founded in 1996 by Anniina Jokinen. Initially, Jokinen created the site as a starting point for

students and enthusiasts of English literature. The site grew, adding links to similar sites such as Renaissance Editions and eventually hosting content shared by colleagues. In essence, the owners of Internet e-book libraries began to do cooperative collection development, often with great success. The process was so easy as to be almost transparent: hyperlink to the other libraries. In June 1993, Lynn H. Nelson launched CARRIE, the first full-text history library on the Web. In a brief article titled "CARRIE: A Full-Text Online Library," Nelson gives a firsthand account of the site's launch, which he indicates was an immediate success. Within a month the site had approximately three thousand links to historical texts.

Despite its success, CARRIE was soon overtaken by other sites. As Nelson writes, "It was only two months until a new day dawned and colleges and universities throughout the country were scrambling to put up World Wide Web sites, and HNSource and CARRIE lost their uniqueness and were overshadowed by the well-funded and professionally staffed projects that began appearing." Libraries and well-funded projects such as the Library of Congress American Memory project, which grew from a pilot digitization project that ran from 1990 to 1994, were uniquely positioned to develop Internet e-book collections. Aside from their obvious expertise in



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Page 78 of 188

- Blended learning: Definition
- Why adopt blended learning?
- Benefits of Blended Learning
- Five Important Considerations Before Adopting a Blended Approach
- Conclusion

Introduction

In the field of second language learning as well as general education, blended and hybrid learning is receiving a lot of attention. Researchers investigating blended learning tend to agree that "blended learning is a transformational force in education." According to Watson (2008), blended learning is "likely to emerge as the predominant model of the future and to become far more common than either [online or face-to-face instruction] alone"

Blended learning as an approach is not new. Indeed, the practice of combining different learning approaches and strategies is not new at all. Distance learning courses have long combined blended learning through a mix of self-access content-print/video/TV/radio-and face-to-face/telephone support.

'Traditional' courses have always combined a variety of delivery modes that combine content such as lectures, seminars, tutorials, workshops and group work to give learners a range of learning opportunities. And of course, 'good teachers will always use more than one method or approach in their teaching, and good learners will always combine different strategies in their learning.'

So we could say that the term 'blended learning' refers to every time teachers mix different media (e.g. print, audio, and video) with classroom interaction, maximizing authentic input to support learners' output and skills development. As such, blended learning has more or less always existed, although the term itself is a mere 15 years old at most, and is now understood to mean a rich, supportive learner-centred learning environment where the 'right blend' is synonymous with effective learning (and teaching).

Page 79 of 188

What is new is that today, the technology combines all the different media within one environment: online. The online space facilitates learner-learner interaction, encourages incidental and exploratory learning and allows learners and teachers to stay connected outside the classroom, if they so wish. Learners can benefit from the fact that space and distance do not